Sneha Mondal

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Summary

Data engineer with 3+ years building and running production data platforms end to end: API ingestion, orchestration, warehouse modeling, transformation as code and data quality, as lead data engineer on a six-tenant platform on Google Cloud. Owned it in production, including monitoring, root-cause debugging, reliability work and a **96%** cut in monthly cloud spend.

Experience

Lead Analytics Engineer (Data Engineering) Jan 2025 – Sep 2026
YNK Incubator (3PL and umbrella company for Veiled Collection + DTC brands) *Elmwood Park, NJ*

- Built and owned the multi-tenant data platform for **six** DTC brands: event-driven ingestion on Cloud Run, Pub/Sub, Eventarc and Cloud Tasks into a BigQuery warehouse per client, with tenant separation enforced at the IAM layer (per-tenant service accounts, least-privilege roles, scoped delegation) on a shared base with brand-specific logic.
- Orchestrated scheduled and event-driven pipelines with dependency ordering, backoff retries and chunked, checkpointed backfills, so a failed run resumed rather than restarting. Modeled transformations as version-controlled SQL in Dataform (dependency graph, assertions, incremental models, CI/CD), so every change was tested before reaching production data.
- Grew Sheets and Apps Script tooling into a client portal (Vercel, Neon Postgres) with contract-first, fail-loud validation so bad data failed at the boundary; trained the new team member on how the platform and the APIs worked.

Lead Data & Analytics Engineer Feb 2023 – Sep 2026
Veiled Collection (same employer group, scope expanded to the YNK platform above in 2025) *Clifton, NJ*

- Built the original single-brand platform (Shopify API into append-only tables, MERGE into current-state models) and cut operational data errors **95%** across **44K SKUs** with validation gates, daily snapshots, snapshot-diff reconciliation and alerting, so buying decisions ran on counts that tied back to source. Synced multi-location inventory against a SKU, barcode and bin grid.
- Built the Buy Plan forecasting service: sell-out dates, reorder points and **6-month** order quantities by size and color, which the buying team planned purchasing against, validated against actual sell-through and corrected where projections drifted. Modeled order geography for the analysis behind the brand's first New York store.
- Reconciled fulfillment status across Shopify, ShipStation and warehouse operations so a stalled order raised itself as an exception, automating what carrier APIs exposed reliably and building scanner and exception-check workflows where they did not.
- Diagnosed a runaway Eventarc to Cloud Run feedback loop and cut monthly GCP spend **96%**, roughly **$8,000 to $300**: event-payload gating, deterministic task names against duplicate work, minimum instances to zero, partitioning and clustering. Architectural, so the saving recurs.

Technical Skills

Languages	Python (pipelines, Pandas, NumPy, pytest), SQL (CTEs, window functions, incremental models), JavaScript/TypeScript, R, Java
Cloud & warehouse	GCP: BigQuery (3 yrs production, partitioning, clustering, cost tuning), Cloud Run, Cloud Functions, Pub/Sub, Eventarc, Cloud Tasks, Scheduler. Snowflake: data loading, table and view design, analytical SQL, warehouse and role configuration. PostgreSQL (Neon), MySQL, SQL Server, MongoDB
Transformation	Dataform in production (version-controlled SQL, dependency graph, assertions, incremental materializations, CI/CD); dbt and DuckDB at project scale; Airflow; dependency ordering, backoff retries, checkpointed backfills, idempotent replay; Docker, Git, CI/CD
Quality & APIs	Data contracts, fail-loud validation, snapshot-diff reconciliation, freshness and row-count checks, monitoring and alerting, root-cause debugging, lineage; REST, GraphQL, Shopify Admin API, webhooks, ShipStation, cursor pagination, rate limits
Analytics	Looker, Tableau, Power BI; dimensional modeling, incremental ELT, SCD Type 2; scikit-learn, NLP (BERT, LSTM); Claude Code and Cursor daily for development and data validation

Projects

- **Data Reliability Pipeline & Analytics Agent** github.com/snehamondal-data/marketing-data-reliability-pipeline
  Python, dbt, DuckDB, Anthropic API. Idempotent incremental ingestion from a rate-limited API (Retry-After backoff, watermark checkpoints, append-only loads, so a rerun inserts **zero** rows) into multi-tenant dbt marts: **84** dbt tests per tenant, **46** pytest. An LLM agent over those marts adds parsed SELECT-only guardrails (tenant-scoped, row and turn caps, logged) and a 10-question eval harness scoring **10/10** live.

Education

Rochester Institute of Technology, M.S. Data Science, 2025 **University of Calcutta**, B.S. (Honors) Mathematics, 2021